

Study Notes on Korteweg (2010): *The Net Benefits to Leverage*

Notes (models, derivations, and results interpretation)

Contents

1	Executive roadmap (what I should be able to explain after reading)	3
2	Core accounting and asset-pricing identities	3
2.1	Value identity and “net benefits” decomposition	3
2.2	Beta definition	4
2.3	Portfolio beta is a value-weighted average: a short proof	4
2.4	Firm beta from the debt–equity identity	4
2.5	Firm beta from the unlevered–net-benefits identity	4
2.6	The key system: equating the two beta decompositions	4
2.7	Special case: only corporate tax shields	5
3	Identification: why assumptions are necessary and what they are	5
3.1	Why the system is underidentified without structure	5
3.2	Assumption (A1): homogeneous unlevered beta within industry	5
3.3	Assumption (A2): net benefits are a function of observables	5
3.4	Figure 1 intuition: why leverage dispersion matters	5
4	Empirical specification: all core models	6
4.1	Leverage definition (used throughout)	6
4.2	Net benefits functional form	6
4.3	Measurement equations (values and betas) with error terms	6
4.4	Return equations (CAPM regressions) and beta dynamics	7
4.5	Optimal leverage: derivation and closed-form expression	7
4.6	Gain to refinancing and over/underleverage	8
4.7	Financial policy model: refinancing probability (logit)	8
5	Data construction (what the paper measures and why it matters)	8
5.1	Core measurement challenge: market value and beta of corporate debt	8
5.2	Sample composition and leverage dispersion	8
5.3	Firm characteristics (used in X_{it})	9
6	Results guide: how to read every table and figure	9
6.1	Table I: industry coverage and sample size	9
6.2	Table II: summary statistics and correlations	10
6.3	Table III: parameter estimates for $g(L, X) = B/V^L$	10
6.4	Figure 2: net benefits to leverage curves	12

6.5	Figure 3: optimal leverage as a function of firm characteristics	12
6.6	Table IV: observed vs market-implied optimal leverage	15
6.7	Table V: determinants of over-/underleverage	15
6.8	Table VI: leveraged buyout (LBO) firms	18
6.9	Figure 4: gain to refinancing vs overleverage	18
6.10	Figure 5: refinancing probability as a function of gain	19
6.11	Figure 6: time series of leverage (observed vs optimal)	20
7	Main conclusions we should remember	20

1 Executive roadmap (what I should be able to explain after reading)

The paper proposes a way to *identify and estimate the present value of the net benefits to debt financing* using:

- market values of corporate **debt** and **equity**, and
- their **systematic risk exposures** (CAPM-style betas).

The key object is a *net-benefits function* B_{it} (present value at time t) such that

$$V_{it}^L = V_{it}^U + B_{it},$$

where V^L is the observed firm value (debt + equity) and V^U is the unlevered firm value (counterfactual value without leverage).

Once B_{it} can be estimated as a function of leverage and firm characteristics, the paper:

- recovers a **market-implied optimal leverage ratio** L_{it}^* ,
- compares L_{it}^* to observed leverage L_{it} (under-/overleverage),
- defines the **gain to refinancing** from moving to L^* ,
- tests whether firms refinance when the gain is large (and where refinancing seems blocked).

My mental map:

(Identities) $\Rightarrow$ (Need assumptions) $\Rightarrow$ (Estimate $B(L, X)$) $\Rightarrow$ (Compute L^* , GAIN) $\Rightarrow$ (Interpret tables/figures).

2 Core accounting and asset-pricing identities

2.1 Value identity and “net benefits” decomposition

At time t , define:

- D_t : market value of debt,
- E_t : market value of equity,
- V_t^L : total market value of the levered firm,
- V_t^U : total market value of the unlevered firm (counterfactual),
- B_t : present value of net benefits of leverage.

The paper starts from two identities:

(1) **Levered firm value is the value of its securities.**

$$V_t^L = D_t + E_t. \tag{1}$$

(2) **Levered firm value equals unlevered value plus net benefits.**

$$V_t^L = V_t^U + B_t. \tag{2}$$

Interpretation:

- V_t^U is what the firm’s assets would be worth without leverage.
- B_t is the *present value* of the difference between benefits and costs of debt financing (tax shields, agency effects, distress costs, etc.), as priced by the market.

2.2 Beta definition

Let r_t^M be the market return. For any asset X with return r_t^X , define the (conditional) CAPM beta:

$$\beta_t^X \equiv \frac{\text{Cov}(r_t^X, r_t^M)}{\text{Var}(r_t^M)}. \quad (3)$$

2.3 Portfolio beta is a value-weighted average: a short proof

Claim. If a portfolio has value $V = V_1 + V_2$ and return $r = w_1 r_1 + w_2 r_2$ with $w_j = V_j/V$, then

$$\beta = w_1 \beta_1 + w_2 \beta_2.$$

Proof.

$$\text{Cov}(r, r^M) = \text{Cov}(w_1 r_1 + w_2 r_2, r^M) = w_1 \text{Cov}(r_1, r^M) + w_2 \text{Cov}(r_2, r^M).$$

Divide both sides by $\text{Var}(r^M)$:

$$\beta = \frac{\text{Cov}(r, r^M)}{\text{Var}(r^M)} = w_1 \frac{\text{Cov}(r_1, r^M)}{\text{Var}(r^M)} + w_2 \frac{\text{Cov}(r_2, r^M)}{\text{Var}(r^M)} = w_1 \beta_1 + w_2 \beta_2.$$

□

2.4 Firm beta from the debt–equity identity

Apply the portfolio-beta result to $V_t^L = D_t + E_t$. The levered firm is a portfolio of debt and equity, so:

$$\beta_t^L = \frac{D_t}{V_t^L} \beta_t^D + \frac{E_t}{V_t^L} \beta_t^E. \quad (4)$$

2.5 Firm beta from the unlevered–net-benefits identity

Apply the same logic to $V_t^L = V_t^U + B_t$. Treat B_t as a “net benefits security” with beta β_t^B . Then:

$$\beta_t^L = \frac{V_t^U}{V_t^L} \beta_t^U + \frac{B_t}{V_t^L} \beta_t^B. \quad (5)$$

2.6 The key system: equating the two beta decompositions

Equations (4) and (5) both describe β_t^L . Therefore,

$$\frac{V_t^U}{V_t^L} \beta_t^U + \frac{B_t}{V_t^L} \beta_t^B = \frac{D_t}{V_t^L} \beta_t^D + \frac{E_t}{V_t^L} \beta_t^E. \quad (6)$$

Together, the value identity (2) and the beta balance (6) form the model’s *core measurement system*.

2.7 Special case: only corporate tax shields

If the only net benefit is the corporate tax shield on debt, then:

$$B_t = \tau D_t \quad \Rightarrow \quad V_t^L = V_t^U + \tau D_t.$$

The paper also shows how this implies a familiar unlevering relation (a variant of the MM relation) in this special case. In my own words:

- when B_t is proportional to D_t , the “net benefits security” is tightly linked to debt,
- but the paper’s general approach is strictly broader because it allows B_t to include non-tax effects and to be negative at high leverage.

3 Identification: why assumptions are necessary and what they are

3.1 Why the system is underidentified without structure

At each (i, t) , the objects V_{it}^U , B_{it} , β_{it}^U , β_{it}^B are not directly observed. With N firms and T dates, there are $4NT$ unknown time-varying quantities, but only about $2NT$ identity conditions from values and betas. So the model needs restrictions that reduce the unknowns.

3.2 Assumption (A1): homogeneous unlevered beta within industry

The paper’s first identifying assumption is:

Firms within an industry share the same unlevered asset beta β^U (the paper discusses two practical versions of this restriction).

Intuition:

- Within a narrow industry, operating-asset risk exposures are similar.
- Then, *cross-sectional variation* in observed debt/equity betas and leverage within the industry helps pin down the unobserved B component.

3.3 Assumption (A2): net benefits are a function of observables

The second identifying assumption is:

The ex ante net benefits to leverage are a function of observable variables.

Operationally, this means I can write:

$$\frac{B_{it}}{V_{it}^L} = g(L_{it}, X_{it}; \theta),$$

where X_{it} is a vector of firm characteristics and θ is a finite-dimensional parameter vector.

3.4 Figure 1 intuition: why leverage dispersion matters

A guiding idea is that we do *not* need firms to be optimally levered to identify $g(\cdot)$. Instead we need:

- enough variation in leverage within an industry,

- and a stable mapping from (L, X) to B/V^L .

The paper emphasizes that frictions can make observed leverage deviate from the market-implied optimum, and that the gap between current and optimal leverage can be interpreted through:

- *transaction costs* of adjusting leverage, and
- *costs of being away from the optimum*,

which jointly appear as “friction costs” in the interpretation of $B(L, X)$.

4 Empirical specification: all core models

4.1 Leverage definition (used throughout)

Define leverage as:

$$L_{it} \equiv \frac{\text{net debt value}}{\text{net debt value} + \text{market equity value}},$$

where net debt is market value of debt net of cash (bounded below by zero).

4.2 Net benefits functional form

The paper parameterizes net benefits as a quadratic function of leverage with firm characteristics shifting the level and curvature:

$$\frac{B_{it}}{V_{it}^L} = X'_{0,it}\theta_0 + (X'_{1,it}\theta_1) L_{it} + (X'_{2,it}\theta_2) L_{it}^2. \quad (7)$$

Key design logic:

- The quadratic in L naturally produces an *inverted-U* (benefits rise at low leverage and fall at high leverage).
- Firm characteristics can shift both the linear and quadratic parts, so the peak (optimal leverage) varies across firms.

Specification I vs II. The paper uses two variants:

- **Specification I:** characteristics enter either the linear (benefits) or the quadratic (costs) term, reflecting a benefit-vs-cost interpretation.
- **Specification II (full):** characteristics may enter both terms, allowing richer patterns (e.g., profitability can raise benefits at low leverage but accelerate diminishing returns at high leverage).

4.3 Measurement equations (values and betas) with error terms

To allow for noise, the paper writes the identities with additive errors.

(i) **Value measurement equation.**

$$V_{it}^U + B_{it} = D_{it} + E_{it} + u_{it}. \quad (8)$$

(ii) **Beta measurement equation.** Start from (6) and move the B -term to the right:

$$\frac{V_{it}^U}{V_{it}^L} \beta_{k(i),t}^U = \frac{D_{it}}{V_{it}^L} \beta_{it}^D + \frac{E_{it}}{V_{it}^L} \beta_{it}^E - \frac{B_{it}}{V_{it}^L} \beta_{it}^B + v_{it}. \quad (9)$$

Here $k(i)$ denotes firm i 's industry, so $\beta_{k(i),t}^U$ is the common industry unlevered beta at time t under (A1).

4.4 Return equations (CAPM regressions) and beta dynamics

The model uses return equations to discipline beta estimates:

CAPM-style return regressions. For each security $X \in \{D, E, U, B\}$,

$$r_{it}^X = \alpha^X + \beta_{it}^X r_t^M + \eta_{it}^X. \quad (10)$$

Notes:

- r^D and r^E are observed from market data (bond and stock returns).
- r^U and r^B are returns on latent components V^U and B , implied by estimated value processes.

Industry unlevered beta dynamics (AR(1)). The paper models the (industry-level) unlevered beta as:

$$\beta_{k,t}^U = \phi_0 + \phi_1 \beta_{k,t-1}^U + \epsilon_{k,t}. \quad (11)$$

4.5 Optimal leverage: derivation and closed-form expression

Definition. The *market-implied optimal leverage* L_{it}^* is the leverage ratio that maximizes firm value. Since $V^L = V^U + B$ and V^U is the value of operating assets (not mechanically a function of current financing choice in the reduced-form mapping), maximizing V^L with respect to L is equivalent to maximizing B .

Key simplification. The model estimates $g(L, X) \equiv B/V^L$. Because:

$$V^L = V^U + B \quad \Rightarrow \quad V^L = V^U + g V^L \quad \Rightarrow \quad V^L = \frac{V^U}{1-g}, \quad B = V^L - V^U = V^U \left(\frac{g}{1-g} \right),$$

maximizing B in L is equivalent to maximizing $g(L, X)$, since $\frac{g}{1-g}$ is increasing in g for $g < 1$.

FOC under quadratic form. Using (7), define

$$g(L, X) = X'_0 \theta_0 + (X'_1 \theta_1) L + (X'_2 \theta_2) L^2.$$

Then:

$$\frac{\partial g}{\partial L} = (X'_1 \theta_1) + 2(X'_2 \theta_2) L.$$

Setting to zero yields the interior optimum:

$$L^*(X) = -\frac{X'_1 \theta_1}{2 X'_2 \theta_2}. \quad (12)$$

In practice, I should remember the feasibility constraint:

$$L^* \in [0, 1] \quad (\text{or } [0, 1 - \varepsilon] \text{ if needed}).$$

4.6 Gain to refinancing and over/underleverage

Overleverage.

$$\text{Overleverage}_{it} \equiv L_{it} - L_{it}^*.$$

Positive values mean the firm is overlevered relative to the model-implied optimum.

Gain to refinancing. Define:

$$\text{GAIN}_{it} \equiv g(L_{it}^*, X_{it}) - g(L_{it}, X_{it}) = \left(\frac{B}{V^L} \right) \Big|_{L=L^*} - \left(\frac{B}{V^L} \right) \Big|_{L=L_{\text{obs}}}. \quad (13)$$

GAIN_{it} is interpreted as the increase in firm value (as a fraction of V^L) if the firm could move costlessly to L^* at time t .

4.7 Financial policy model: refinancing probability (logit)

For underlevered firms, the paper models the probability of a major leverage-increasing refinancing event in year $t + 1$ as:

$$\ln \left(\frac{p_{i,t+1}}{1 - p_{i,t+1}} \right) = \beta_0 + \beta_1 \text{GAIN}_{it} + \beta_2 \text{GAIN}_{it}^2 + \beta_3 \text{GAIN}_{it}^3 + \eta_{it}. \quad (14)$$

A symmetric logit is estimated for overlevered firms (major leverage-decreasing actions), using the same functional form in GAIN.

5 Data construction (what the paper measures and why it matters)

5.1 Core measurement challenge: market value and beta of corporate debt

A central empirical requirement is to measure:

$$D_{it} \quad \text{and} \quad r_{it}^D \Rightarrow \beta_{it}^D.$$

The paper uses traded bond prices/transactions and bond characteristics to build firm-level debt values and debt returns. I should remember two facts:

- Only a fraction of total debt is directly observed in bond markets each month.
- The remainder (e.g., bank debt) is proxied (the paper discusses how to handle the “unobserved” part).

5.2 Sample composition and leverage dispersion

The sample is constructed to ensure enough leverage variation within industries, which is critical for identification.

Table I
Sample Breakdown by Industry

This table presents a breakdown of the 1994 to 2004 sample into industries defined by two-digit SIC codes (SIC2, in Panel A) or Fama–French (1997) classification (FF, in Panel B). The observed number of firms (# Firms) and firm-months (N) are reported for each industry.

Panel A: SIC2			
Name	Code	# Firms	N
Oil & gas	13	11	684
Builders	15	5	504
Food	20	12	1,416
Paper	26	6	768
Publishing	27	7	827
Chemicals	28	24	2,844
Petroleum products	29	4	480
Primary metals	33	5	480
Machinery	35	16	1,572
Electric equipment	36	26	2,275
Cars	37	8	972
Instruments	38	5	648

5.3 Firm characteristics (used in X_{it})

The key firm characteristics used in Table II and the net-benefits specification are:

- **PROF:** EBITDA / Sales.
- **DEPR:** Depreciation and amortization / Total assets.
- **VOL:** Volatility of profitability growth (as defined in the paper’s Table II note).
- **PPE:** Net property, plant, and equipment / Total assets.
- **MB:** Market-to-book ratio of equity (as defined in the paper’s Table II note).
- **LN(TA):** $\ln(\text{total assets})$.
- **D_RECESS:** recession dummy (NBER peak-to-trough).

6 Results guide: how to read every table and figure

This section is my interpretation checklist: what each table/figure contains, how it is constructed, and what I should take away.

6.1 Table I: industry coverage and sample size

What it is. Table I lists industries and reports # firms and # firm-months for:

- the SIC2 industry partition, and
- the Fama–French (1997) industry partition.

How I read it.

- The table reveals that the sample is not “all public firms”; it is a bond-availability and measurement-driven sample.
- The paper explicitly wants enough within-industry leverage dispersion for identification.

Core takeaway. This table is a sanity check that the panel is large enough and spread across industries so that (A1) is plausible and informative.

Table II
Summary Statistics

This table presents summary statistics for the 29,753 firm-month observations in the FF sample. Variables are defined as follows: *PROF* is the most recent EBITDA/Sales (annual Compustat item 13 divided by item 12); *DEPR* is depreciation over book assets (item 14/item 6); *VOL* is the standard deviation of $(PROF_{i,t+1} - PROF_{it})/PROF_{it}$; *PPE* is property, plant, and equipment divided by book assets (item 8/item 6); *MB* is equity market capitalization divided by book equity (item 6 minus item 181); *LN(TA)* is the natural logarithm of total assets (item 6, in millions); and *L* is the net debt value (market value of debt net of cash) divided by the sum of net debt and market value of equity, bounded below by zero. The book value of the unobserved portion of debt is used as a proxy for its market value. *SD* is the standard deviation. Source: FISD, CRSP, and Compustat.

Panel A: Summary Statistics					
	Mean	Percentile			<i>SD</i>
		10	50	90	
PROF	0.174	0.045	0.157	0.338	0.123
DEPR	0.044	0.015	0.040	0.073	0.033
VOL	0.120	0.017	0.045	0.274	0.238
PPE	0.328	0.055	0.289	0.663	0.229
MB	1.739	0.307	1.014	3.717	3.044
LN(TA)	7.508	4.968	7.655	9.684	1.859
<i>L</i>	0.234	0	0.179	0.573	0.233

Panel B: Correlation Matrix						
	DEPR	VOL	PPE	MB	LN(TA)	<i>L</i>
PROF	0.185	-0.189	0.167	0.091	0.189	0.002
DEPR		-0.067	0.404	-0.043	-0.060	0.069
VOL			-0.129	0.181	-0.272	-0.126
PPE				-0.172	0.178	0.333
MB					-0.163	-0.346
LN(TA)						0.258

6.2 Table II: summary statistics and correlations

What it is. Table II reports summary statistics for the main variables and a correlation matrix.

How I read Panel A (distributions). I use Panel A to learn:

- the typical leverage level in the sample,
- the scale and dispersion of profitability, volatility, tangibility, and market-to-book,
- whether leverage has a mass at or near zero (important for the “zero leverage” discussion later).

How I read Panel B (correlations). I use Panel B to compare:

- the raw leverage correlations (reduced-form),
- against the paper’s market-implied optimal-leverage patterns (structural interpretation).

Core takeaway. Raw correlations are informative but can be misleading if firms are not at L^* . The paper’s approach aims to be robust to this issue by identifying the entire $g(L, X)$ curve.

6.3 Table III: parameter estimates for $g(L, X) = B/V^L$

What it is. Table III provides posterior means and uncertainty measures for the parameter vectors $\theta_0, \theta_1, \theta_2$ in (7).

Table III
Parameter Estimates

This table reports the posterior mean and standard deviation of parameter estimates over the 1994 to 2004 sample, defining industries either by their two-digit SIC code (SIC2) or the Fama–French (1997) industry classifications (FF). The model specifies the net benefits to leverage relative to firm value (B/V^L) as a function of firm characteristics:

$$B_{it}/V_{it}^L = X_{0it}^L \theta_0 + (X_{1it}^L \cdot L_{it}) \theta_1 + (X_{2it}^L \cdot L_{it}^2) \theta_2.$$

The explanatory variables are as defined in Table II. The dummy variable *D.RECESS* equals one in a recession, as defined by NBER peak-to-trough periods, and zero otherwise. Standard errors are in parentheses. ***, **, and * denote parameter estimates for which zero falls outside the 99%, 95%, and 90% posterior confidence intervals, respectively. *N* is the number of firm-months, and *MSE* is the mean squared error of model residuals.

	I		II	
	SIC2	FF	SIC2	FF
Constant	0.030 (0.009)***	0.041 (0.012)***	−0.013 (0.003)***	−0.012 (0.004)***
PROF	.	.	0.126 (0.005)***	0.150 (0.008)***
DEPR	.	.	−0.301 (0.046)***	−0.288 (0.034)***
VOL	.	.	−0.066 (0.006)***	−0.075 (0.003)***
PPE	.	.	0.110 (0.023)***	0.078 (0.010)***
MB	.	.	−0.001 (0.000)***	−0.001 (0.000)***
D.RECESS	.	.	−0.003 (0.002)	0.004 (0.002)**
LN(TA)	.	.	0.001 (0.001)	0.001 (0.001)
<i>L</i> *				
Constant	0.116 (0.013)***	0.140 (0.020)***	0.171 (0.008)***	0.143 (0.018)***
PROF	0.218 (0.008)***	0.227 (0.015)***	0.415 (0.010)***	0.539 (0.014)***
DEPR	−0.531 (0.030)***	−0.574 (0.027)***	−1.970 (0.287)***	−1.940 (0.116)***
VOL	−0.103 (0.010)***	−0.095 (0.004)***	−0.106 (0.006)***	−0.086 (0.004)***
PPE	.	.	0.243 (0.025)***	0.353 (0.064)***
MB	.	.	−0.001 (0.001)	0.002 (0.005)
D.RECESS	.	.	−0.035 (0.003)***	−0.037 (0.005)***
LN(TA)	0.008 (0.003)***	0.003 (0.004)	−0.002 (0.001)***	−0.003 (0.001)**
<i>L</i> ² *				
Constant	−0.193 (0.006)***	−0.214 (0.006)***	−0.343 (0.040)***	−0.346 (0.052)

(continued)

Table III—Continued

	I		II	
	SIC2	FF	SIC2	FF
PROF	.	.	−0.202 (0.071)***	−0.203 (0.053)***
DEPR	.	.	1.679 (0.041)***	1.655 (0.059)***
VOL	.	.	−0.098 (0.004)***	−0.184 (0.004)***
PPE	0.164 (0.015)***	0.198 (0.010)***	0.124 (0.012)***	0.149 (0.018)***
MB	−0.096 (0.001)***	−0.122 (0.001)***	−0.060 (0.003)***	−0.063 (0.003)***
D_RECESS	−0.060 (0.004)***	−0.079 (0.015)***	−0.048 (0.002)***	−0.052 (0.003)***
LN(TA)	−0.000 (0.001)	0.001 (0.000)***	−0.001 (0.001)	−0.006 (0.001)***
<i>N</i>	24,277	29,753	24,277	29,753
MSE	0.0358	0.0345	0.0241	0.0276

How I read it.

- The sign of the linear-in- L component suggests how benefits rise at low leverage.
- The sign of the quadratic term determines whether net benefits eventually decline (inverted-U).
- How characteristics load into both terms tells me how the *peak* of the curve moves with X .

Core takeaway. The estimation implies a well-defined interior optimum for many firms: net benefits increase at low leverage but decrease at high leverage, consistent with a value-maximizing L^* .

6.4 Figure 2: net benefits to leverage curves

What it is. Figure 2 plots $g(L, X)$ as a function of L under different values of a single characteristic, keeping others fixed.

How I interpret it. For each panel (PROF, DEPR, VOL, PPE, MB, LN(TA), and recession):

- the curve’s peak is the optimal leverage L^* ,
- shifts up/down mean “more/less total net benefits”,
- shifts in the peak location mean “higher/lower optimal leverage”.

Core takeaway. The figure visually confirms the estimated inverted-U shape and demonstrates how firm characteristics tilt the curve and move the optimum.

6.5 Figure 3: optimal leverage as a function of firm characteristics

What it is. Figure 3 plots $L^*(X)$ from (12) as one characteristic varies across its empirical range.

How I interpret it. The figure answers: “If two firms differ only by characteristic X_j , how does the model-implied L^* differ?” I read it as a *comparative statics* diagram.

Core takeaway. The paper highlights a notable pattern: optimal leverage increases with profitability and decreases with firm size, contrasting with many reduced-form regressions.

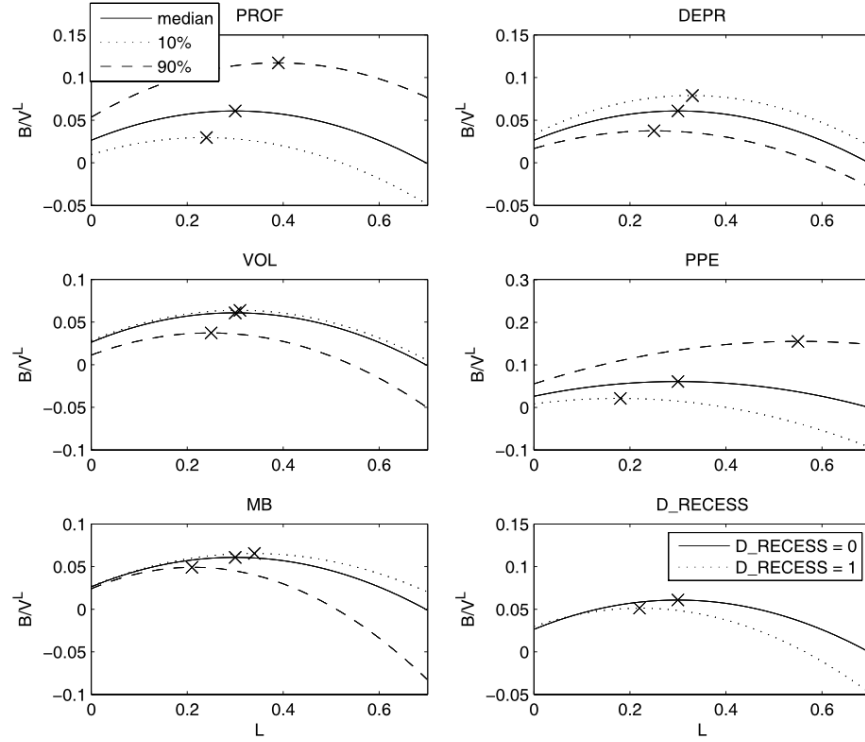


Figure 2. Net benefits to leverage. This figure shows how net benefits to leverage, as a fraction of total firm value (B/V^L), varies with leverage (L , on the horizontal axis). The plots are based on parameter estimates of the full model (Specification II) and using the FF industries, as reported in Table III. The graphs compare the median firm (in an economic expansion) to the same firm but with one characteristic at either its 10th or 90th percentile of the sample distribution. Firm characteristics are as defined in Table II. The bottom-right graph shows the net benefits for a firm with median characteristics in an expansion period versus in a recession (peak-to-trough as defined by the NBER). Optimal leverage ratios, as determined by maximum net benefits, are marked with an "X."

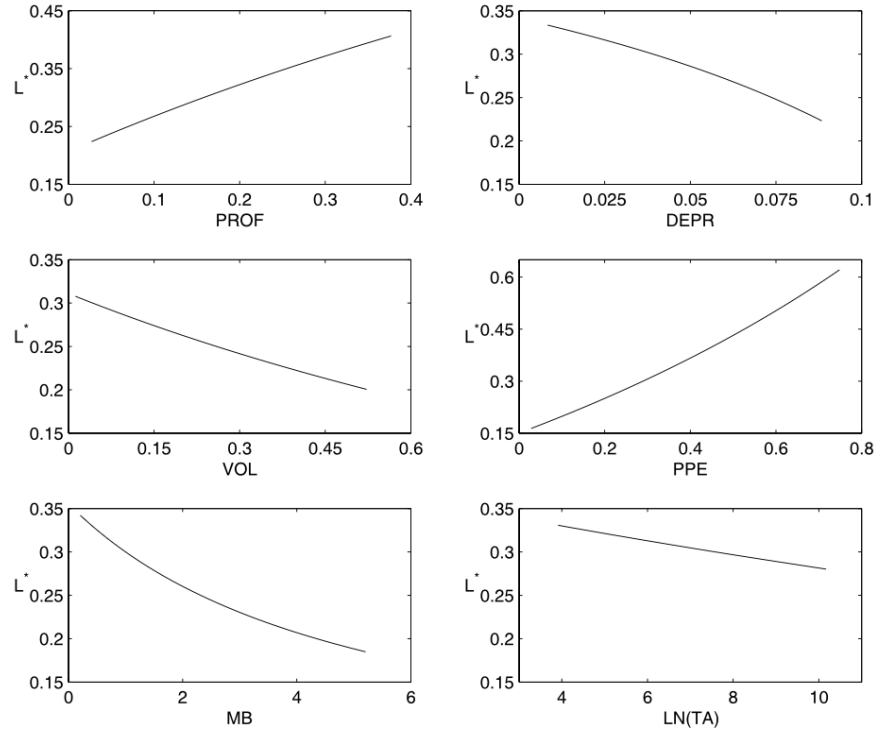


Figure 3. Optimal capital structure. This figure shows how optimal leverage (L^*) varies with firm characteristics (on the horizontal axis), based on parameter estimates of the full model (Specification II) and the FF sample, as reported in Table III. Firm characteristics are as defined in Table II. Each plot allows one firm characteristic to range between the 5th to 95th percentile of the observed data, while keeping the other characteristics at the median of the sample distribution.

Table IV
Optimal versus Observed Leverage

Panels A and B show summary statistics of observed and model-implied optimal leverage for the two-digit SIC (SIC2) and Fama–French (FF) samples, respectively, where leverage is as defined in Table II. Optimal leverage is calculated for each firm-month from Specification II in Table III. Each sample is split into subsamples of interest-paying firms, non-interest-paying firms, and firms that pay neither interest nor dividends. N is the number of firm-months in each subsample.

Panel A: SIC2 Sample					
		Percentile			
	Mean	10	50	90	SD
Full sample ($N = 24,277$)					
Observed	0.240	0	0.196	0.563	0.226
Optimal	0.258	0.128	0.221	0.432	0.152
Interest-paying firms ($N = 22,595$)					
Observed	0.256	0	0.227	0.575	0.225
Optimal	0.264	0.133	0.224	0.441	0.154
Non-interest-paying firms ($N = 1,682$)					
Observed	0	0	0	0	0
Optimal	0.180	0.042	0.196	0.258	0.088
Non-interest, non-dividend-paying firms ($N = 1,060$)					
Observed	0	0	0	0	0
Optimal	0.150	0	0.163	0.247	0.091
Panel B: FF Sample					
Full sample ($N = 29,753$)					
Observed	0.234	0	0.179	0.573	0.233
Optimal	0.316	0.142	0.269	0.575	0.176
Interest-paying firms ($N = 26,868$)					
Observed	0.259	0	0.225	0.591	0.232
Optimal	0.328	0.155	0.279	0.591	0.176
Non-interest-paying firms ($N = 2,885$)					
Observed	0	0	0	0	0
Optimal	0.205	0.065	0.189	0.379	0.127
Non-interest, non-dividend-paying firms ($N = 1,972$)					
Observed	0	0	0	0	0
Optimal	0.163	0	0.159	0.289	0.100

6.6 Table IV: observed vs market-implied optimal leverage

What it is. Table IV reports percentiles of observed leverage and optimal leverage, for both models (SIC2 and FF) and both specifications.

How I read it.

- Compare the median observed leverage to the median optimal leverage: under- vs overleverage at the center.
- Compare tails: do high-leverage firms appear overlevered? do low-leverage firms appear underlevered?
- Look at subsamples: interest-paying vs (true) zero-leverage firms.

Core takeaway. The median firm appears underlevered, but the extent depends heavily on whether true zero-leverage firms are included. This table sets up the “zero leverage puzzle”.

6.7 Table V: determinants of over-/underleverage

Table V
Drivers of Over- and Underleverage

This table presents OLS regressions of observed minus model-implied optimal leverage for the two-digit SIC (SIC2) and Fama–French (FF) samples, respectively. Leverage (L) is as defined in Table II. The dummy variables D_INT0 and D_DIV0 equal one when a firm does not pay interest or dividends, respectively, in that month. The dummy variable D_NONINV equals one if a firm is rated below investment grade in that month. Other variables are as explained in Table II. Specification III includes both firm and year fixed effects. Standard errors are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Dependent = Observed L – Optimal L						
	SIC2			FF		
	I	II	III	I	II	III
D.INT0	−0.143 (0.007)***	−0.126 (0.008)***	0.017 (0.010)*	−0.094 (0.005)***	−0.016 (0.005)***	0.015 (0.012)
D.INT0 · D.DIV0	0.081 (0.011)***	0.039 (0.009)***	0.005 (0.022)	0.133 (0.009)***	0.088 (0.008)***	−0.028 (0.012)**
D.NONINV	.	0.177 (0.003)***	0.021 (0.003)***	.	0.164 (0.003)***	0.048 (0.003)***
D.RECESS	.	0.101 (0.005)***	0.071 (0.006)***	.	0.124 (0.004)***	0.083 (0.005)***
PROF	.	−0.373 (0.010)***	−0.481 (0.014)***	.	−0.510 (0.009)***	−0.540 (0.011)***
MB	.	0.008 (0.000)***	0.009 (0.000)***	.	−0.027 (0.001)***	−0.012 (0.001)***
LN(TA)	.	0.015 (0.001)***	0.011 (0.002)***	.	0.012 (0.001)***	0.001 (0.002)

(continued)

Table V—Continued

Dependent = Observed L – Optimal L						
	SIC2			FF		
	I	II	III	I	II	III
Intercept	−0.006 (0.001)***	−0.120 (0.007)***	0.026 (0.016)***	−0.069 (0.001)***	−0.085 (0.006)***	−0.127 (0.016)***
Firm-FE	N	N	Y	N	N	Y
Year-FE	N	N	Y	N	N	Y
Adjusted- R^2	0.044	0.253	0.698	0.040	0.312	0.773
F	559.743	1175.170	227.476	617.633	1929.622	330.715
p	0.000	0.000	0.000	0.000	0.000	0.000
N	24,277	24,277	24,277	29,753	29,753	29,753

Table VI
Leveraged Buy-Out Firms

This table presents firm characteristics and model-implied optimal leverage ratios for 25 leveraged buy-out firms from Andrade and Kaplan (1998), using accounting data from Compustat in the fiscal year before the buy-out. HLT Date is the date of the buyout. Leverage (L) is the book value of debt (net of cash) relative to net debt plus market equity, calculated from Andrade and Kaplan. Firm characteristics are defined in Table II. I report leverage both in the year before (Pre) and after ($Post$) the buy-out. Optimal leverage (Opt) is calculated from Specification II in Table III for both the SIC2 and FF models. The gain to refinancing ($GAIN$) is calculated as B_{it}/V_{it}^L at the optimal leverage minus B_{it}/V_{it}^L at the pre-buy-out leverage.

Name	HLT Date	PROF	DEPR	VOL	PPE	MB	LN(TA)	L		SIC2		FF	
								Pre	Post	Opt	GAIN	Opt	GAIN
Burlington Inds	9/87	0.092	0.063	0.209	0.496	0.523	7.688	0.107	1.007	0.532	0.046	0.345	0.017
Cherokee	5/89	0.143	0.014	0.183	0.130	1.055	4.526	−0.033	0.417	0.277	0.040	0.233	0.033
Florida Steel	11/88	0.123	0.042	0.365	0.549	0.619	5.634	0.198	1.058	0.546	0.037	0.387	0.013
Fort Howard	10/88	0.253	0.057	0.144	0.582	1.059	7.694	0.113	0.903	0.632	0.084	0.487	0.049
Harcourt Brace Jovanovich	7/87	0.178	0.031	0.147	0.325	0.625	7.444	0.235	0.965	0.432	0.013	0.336	0.004
Hills Stores	12/85	0.087	0.037	0.200	0.350	0.162	6.640	0.169	0.839	0.442	0.022	0.310	0.007
Interco	12/88	0.096	0.032	0.100	0.241	0.588	7.594	0.161	0.891	0.348	0.012	0.254	0.003
KDI	12/88	0.100	0.036	0.176	0.326	0.581	5.212	0.205	1.045	0.405	0.013	0.301	0.003
Leaseway Transp	6/87	0.097	0.094	0.166	0.558	0.507	6.820	0.245	0.475	0.649	0.031	0.391	0.005
R H Macy	7/86	0.122	0.044	0.084	0.464	1.356	7.765	0.265	0.855	0.477	0.015	0.351	0.003
Mayflower	12/86	0.070	0.056	0.210	0.467	0.663	5.547	0.307	0.917	0.490	0.009	0.331	0.000
National Gypsum	4/86	0.198	0.040	0.374	0.453	0.953	6.957	0.030	0.850	0.449	0.063	0.337	0.040
Papercraft	10/85	0.177	0.021	0.142	0.217	1.218	4.968	−0.035	0.795	0.335	0.056	0.282	0.045
Payless Cashways	10/88	0.072	0.043	0.105	0.522	0.475	6.703	0.221	0.901	0.612	0.041	0.429	0.013
Pay 'N' Pak	3/88	0.057	0.024	0.080	0.455	0.361	5.847	0.248	0.968	0.565	0.030	0.415	0.009
Plantronics	4/89	0.187	0.039	0.138	0.171	1.160	4.773	0.106	0.976	0.292	0.013	0.246	0.008
Republic Health	8/86	0.210	0.032	0.190	0.634	0.229	6.724	0.554	0.924	0.764	0.013	0.585	0.000
Revo	12/86	0.058	0.035	0.237	0.305	0.837	6.895	0.171	1.041	0.331	0.009	0.228	0.001
RJR Nabisco	3/89	0.205	0.035	0.055	0.346	1.158	9.784	0.133	0.923	0.431	0.033	0.336	0.017

(continued)

Table VI—Continued

Name	HLT Date	PROF	DEPR	VOL	PPE	MB	LN(TA)	L		SIC2		FF	
								Pre	Post	Opt	GAIN	Opt	GAIN
Seaman Furniture	2/88	0.145	0.020	0.200	0.379	2.805	4.625	0.013	0.943	0.340	0.052	0.279	0.037
Specialty Eqpm	9/88	0.174	0.044	0.200	0.161	0.464	5.393	0.231	0.961	0.292	0.001	0.234	0.000
Southland	12/87	0.052	0.054	0.122	0.664	0.664	8.138	0.267	0.897	0.739	0.054	0.485	0.013
Supermarkets Gnrl	10/87	0.041	0.066	0.040	0.540	0.765	7.067	0.081	0.911	0.612	0.065	0.398	0.026
USG	7/88	0.184	0.046	0.385	0.568	0.718	7.647	0.265	1.025	0.554	0.027	0.396	0.007
Jim Walter	1/88	0.182	0.036	0.153	0.313	0.818	7.983	0.213	1.013	0.402	0.013	0.311	0.004
Mean		0.132	0.042	0.176	0.409	0.815	6.643	0.179	0.900	0.478	0.032	0.347	0.014
Median		0.123	0.039	0.166	0.453	0.664	6.820	0.198	0.923	0.449	0.030	0.336	0.008
SD		0.060	0.017	0.090	0.153	0.517	1.317	0.124	0.153	0.139	0.022	0.089	0.015

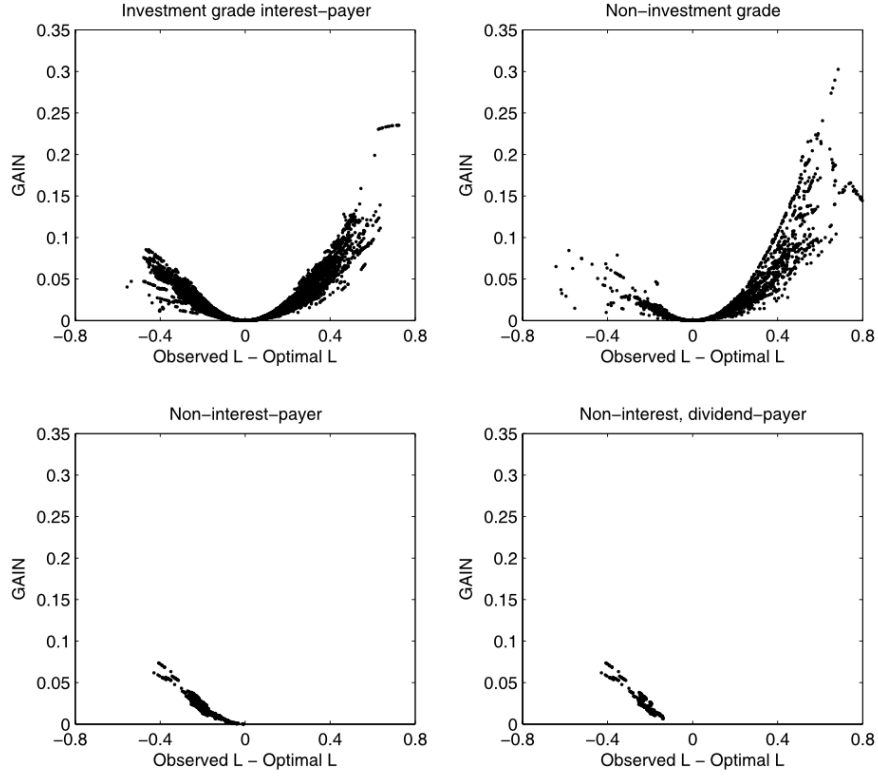


Figure 4. Gain to refinancing. This figure shows plots of the gain to refinancing from observed leverage to the model-implied optimal leverage versus the degree of overleveraging, for the 29,753 firm-month observations in the FF model. Overleveraging is defined as observed leverage minus optimal leverage. Observed leverage (L) is the net debt value (market value of debt net of cash) divided by the sum of net debt and market value of equity. Optimal leverage is calculated from the full model (Specification II) using the FF model, as reported in Table III. The gain to refinancing (GAIN) is calculated as the difference in net benefits (B/V^L) at the optimal leverage minus B/V^L at the observed leverage ratio. The sample is split into firm-years in which firms were rated investment grade and paid interest (top left), firm-years in which firms were rated below investment grade (top right), firm-years in which firms paid no interest (bottom left), and firm-years in which firms paid no interest but did pay dividends (bottom right).

What it is. Table V runs regressions where the dependent variable is:

$$L_{it} - L_{it}^*.$$

Key regressors. The table includes dummies capturing financing constraints/distress:

- D_INT0 : no interest payments,
- D_DIV0 : no dividends,
- D_NONINV : below investment grade,
- D_RECESS : recession.

How I interpret signs.

- A positive coefficient means that variable is associated with *higher overleverage* (observed – optimal).
- A negative coefficient means the variable is associated with *more underleverage*.

Core takeaway. Consistent with intuition, distress and recessions are associated with higher overleverage. Zero-leverage dummies become much less informative after firm fixed effects, suggesting persistence.

6.8 Table VI: leveraged buyout (LBO) firms

What it is. Table VI lists 25 LBO firms and reports:

- firm characteristics pre-buyout,
- leverage pre and post buyout,
- model-implied L^* ,
- the gain to leveraging up from the pre-buyout leverage.

How I interpret it. This table provides an external “case study” on whether LBO targets look underlevered relative to the market-implied optimum.

Core takeaway. LBO targets appear underlevered pre-buyout, but the median gain from leveraging up to L^* is modest relative to the dramatic leverage jump in LBO transactions; post-buyout leverage is far above L^* , consistent with planned debt paydown dynamics.

6.9 Figure 4: gain to refinancing vs overleverage

What it is. Figure 4 plots $GAIN_{it}$ against overleverage ($L - L^*$), and splits the sample into categories (investment grade vs non-investment grade; interest payer vs non-interest payer; dividend payer subset).

How I interpret it.

- The U-shape reflects that being far from L^* is value-relevant.
- Extremely overlevered distressed firms show very large GAIN from delevering, interpreted as large expected distress costs.
- Underlevered firms can also have sizable gains from leveraging up; comparing those gains to issuance costs is an implicit friction-cost diagnostic.

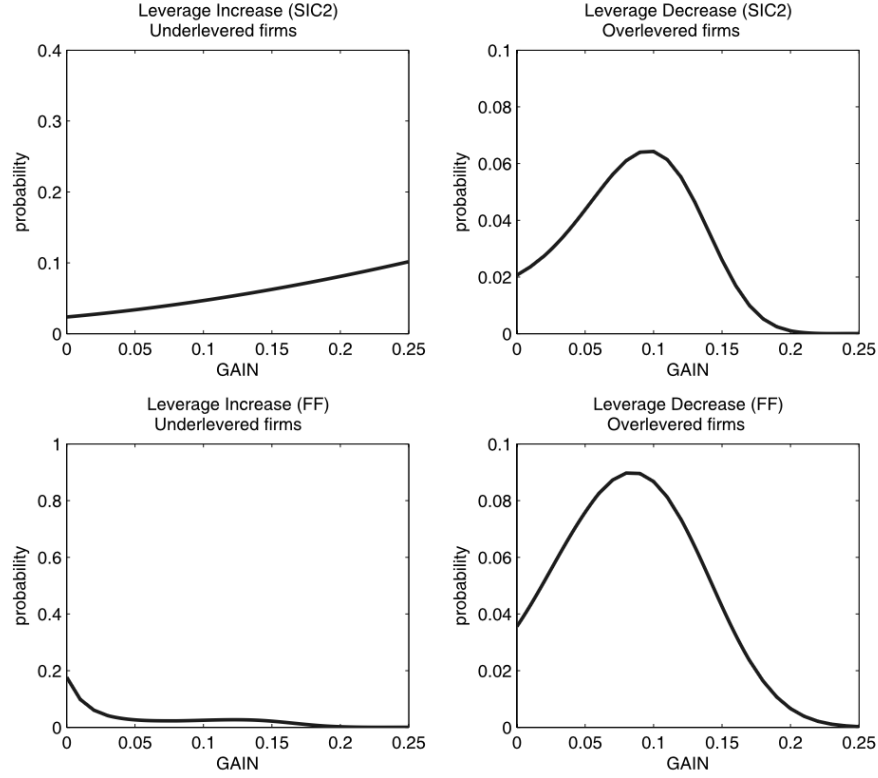


Figure 5. Probability of refinancing. This figure shows plots of the probability of rebalancing the capital structure to the optimum as a function of the gain to refinancing. Underlevered firms are defined as firms with a leverage ratio below (above) the model-implied optimum in a particular year, where optimal leverage is calculated from Specification II in Table III. The probability of levering up for underlevered firms is calculated from a logistic regression:

$$\ln\left(\frac{p_{i,t+1}}{1-p_{i,t+1}}\right) = \beta_0 + \beta_1 * GAIN_{it} + \beta_2 * GAIN_{it}^2 + \beta_3 * GAIN_{it}^3 + \eta_{it},$$

where $p_{i,t+1}$ is the probability of firm i issuing debt or buying back equity worth more than 20% of its book assets in year $t + 1$. The gain to refinancing ($GAIN$) is calculated as in Figure 4. There are 1,213 underlevered firm-years in the SIC2 sample and 1,748 in the FF model. The probability of levering down is calculated similarly for overlevered firm-years, with the probability of firm i issuing equity or buying back debt worth more than 20% of its book value in year $t + 1$ as the dependent variable. There are 743 overlevered firm-years in the SIC2 model, and 656 in the FF model.

6.10 Figure 5: refinancing probability as a function of gain

What it is. Figure 5 plots predicted refinancing probabilities from the logit model (14) for both:

- levering up (underlevered firms), and
- levering down (overlevered firms).

Core reading.

- For underlevered firms, refinancing probability rises with GAIN: firms act more when the value stakes are higher.
- For overlevered firms, the probability can rise initially but then fall for very high GAIN: the firms that *should* delever most are the ones that seem most unable to do so, consistent with market frictions/distress constraints.

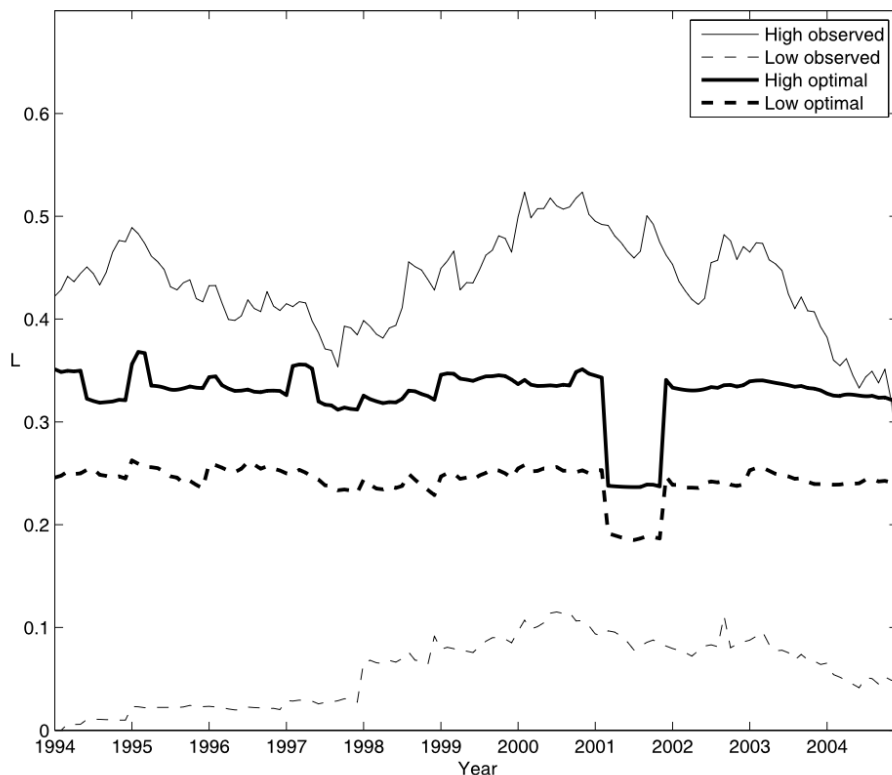


Figure 6. Time series of leverage. This figure shows the observed and optimal leverage for high- and low-leverage portfolios. I sort the 132 firms that exist over the entire 1994 to 2004 sample period into four leverage portfolios, based on initial leverage in January 1994. The plot graphs the average observed leverage ratio of the highest and lowest leverage portfolios, and the associated model-implied optimal leverage from the FF model. Leverage (L) is as defined in Table II.

6.11 Figure 6: time series of leverage (observed vs optimal)

What it is. Figure 6 tracks portfolio-average leverage over time for high- vs low-leverage portfolios, showing both observed and model-implied optimal leverage.

How I interpret it.

- Persistence: observed leverage does not instantly revert to L^* .
- High-leverage portfolio: gradual delevering toward L^* .
- Low-leverage portfolio: slow drift upward but often staying below L^* , driven by persistent zero-leverage firms.

7 Main conclusions we should remember

1. The method identifies net benefits to leverage using $\{D, E, \beta^D, \beta^E\}$ and two assumptions: industry common β^U and $g(L, X)$ depends on observables.
2. Net benefits are hump-shaped in leverage: positive at low leverage and negative at very high leverage, implying an optimal L^* .
3. For the median firm, net benefits at the optimum can be economically meaningful (a few percent of firm value).

4. The median firm appears underlevered, but this is largely driven by true zero-leverage firms.
5. Distressed firms and recessions are associated with overleverage.
6. Refinancing behavior reacts to gains, but the most overlevered distressed firms appear constrained in delevering.